

Ph. D. Examination in Measure and Integration.

May 9, 1988 6 - 10 p.m. Little 125

Instructions: Answer seven out of the eight questions. Justify all work. to obtain any partial credit, all work must be presented in a neat and logical fashion.

Question 1. Prove that $L^1(S, A, \mu)$ is a complete space.

Question 2. State and prove the Hahn-Banach theorem.

Question 3. State and prove Fubini's theorem.

Question 4. Prove that there exists no sequence $(c_n)_{n=1}^{\infty}$ of real numbers such that an infinite series $\sum a_n$ of real numbers converges if and only if $(c_n a_n)_{n=1}^{\infty}$ is a bounded sequence.

[Hint: Without loss of generality, assume such a sequence exists, with $c_n \neq 0$ for all n . Consider the mapping $T: \ell_{\infty} \rightarrow \ell_1$ given by $T((x_n)) = \frac{x_n}{c_n}$ and use the open mapping theorem.]

Question 5. Prove that there exists no Banach space of algebraic dimension $\aleph_0$

[Hint: show that a finite dimensional subspace is closed and then use the Baire category theorem.]

Question 6. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a Lebesgue measurable function. Show that there exists a Borel measurable function g such that $f = g$ a.e. (a.e. with respect to Lebesgue measure).

Question 7. Let μ be a measure on $([0, \infty), \mathcal{B}([0, \infty))$ defined by

$$\mu(A) = \int_A x \, dx .$$

Let $T: [0, \infty) \rightarrow [0, \infty)$ be given by $T(x) = e^x - 1$. Define a new measure π on $([0, \infty), \mathcal{B}([0, \infty))$ by setting $\pi(A) = \mu(T^{-1}(A))$. Compute $d\pi/d\mu$.

Question 8. Let λ be Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. If $A \in \mathcal{B}(\mathbb{R})$, define the set $-A = \{x : -x \in A\}$. Define $\mathcal{C} = \{A \in \mathcal{B}(\mathbb{R}) : A = -A\}$.

Part (a) Show $\mathcal{C}$ is a sigma-algebra.

Part (b) Define two new measures μ and ν on the measurable space $(\mathbb{R}, \mathcal{C})$ by setting

$$\mu(A) = \int_A x \, dx \quad \text{whenever } A \in \mathcal{C}.$$

$$\nu(A) = \int_{A \cap [0, \infty)} x \, dx \quad \text{whenever } A \in \mathcal{C}.$$

Since λ may also be considered as a measure on $(\mathbb{R}, \mathcal{C})$, we can find two $\mathcal{C}$ -measurable functions $d\mu/d\lambda$ and $d\nu/d\lambda$ so that

$$\begin{aligned}\mu(A) &= \int_A d\mu/d\lambda \, dx && \text{whenever } A \in \mathcal{C} \\ \nu(A) &= \int_A d\nu/d\lambda \, dx && \text{whenever } A \in \mathcal{C}\end{aligned}$$

Compute these two functions.